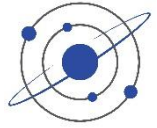


PHY-7012 Assignment #1

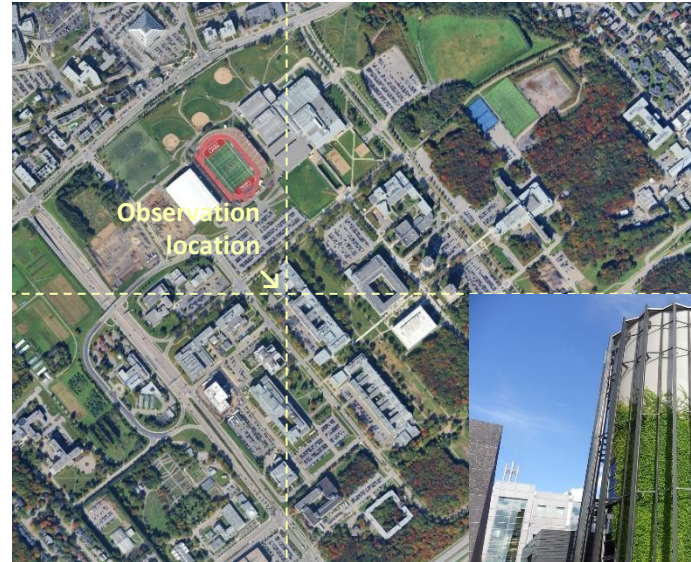
- Due date: Monday, October 12
- This counts for 20% of the final note



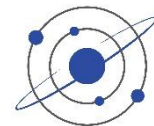


Exercise: ISS passing overhead

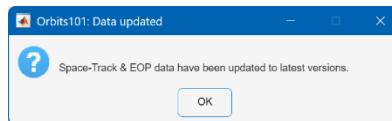
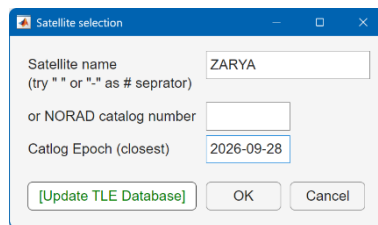
- Simulation for a satellite in overpass over Laval University campus $\varphi = 46.7809^\circ$, $\lambda \equiv -71.2778^\circ$ *
Exact point COPL Department (Centre d'optique, photonique et lasers) <https://copl.fsg.ulaval.ca/>
 - Former Van DeGraff accelerator tower, dismantled and retrofitted to become a high-performance supercomputing hub
 - The “Colosse” is a highly advanced, liquid-cooled supercomputer, directly capturing the massive heat generated by the servers to warm neighboring buildings on campus
 - The facility supports cutting-edge artificial intelligence. The silo houses [tamIA](#), a powerful academic AI computing cluster launched in partnership with Mila and the Digital Research Alliance of Canada
- * Try entering coordinates on Google Earth Pro



Get the satellite orbital elements



- Use the latest COEs describing the ISS orbit from TLE values provided by NORAD
 - You can download manually from the website <https://www.space-track.org/auth/login>
 - Or let Orbits101 download it for you: Pre-defined orbit > NORAD Name/# [Update TLE Database]

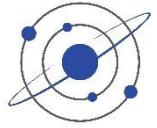


- then find file in:
C:\Program Files\Orbits101\application\SPACE\DATA\Space-Track\Space-Track Full Catalog 3LE (2026-09-28).txt
- Extract COEs from the Zarya (ISS) TLE

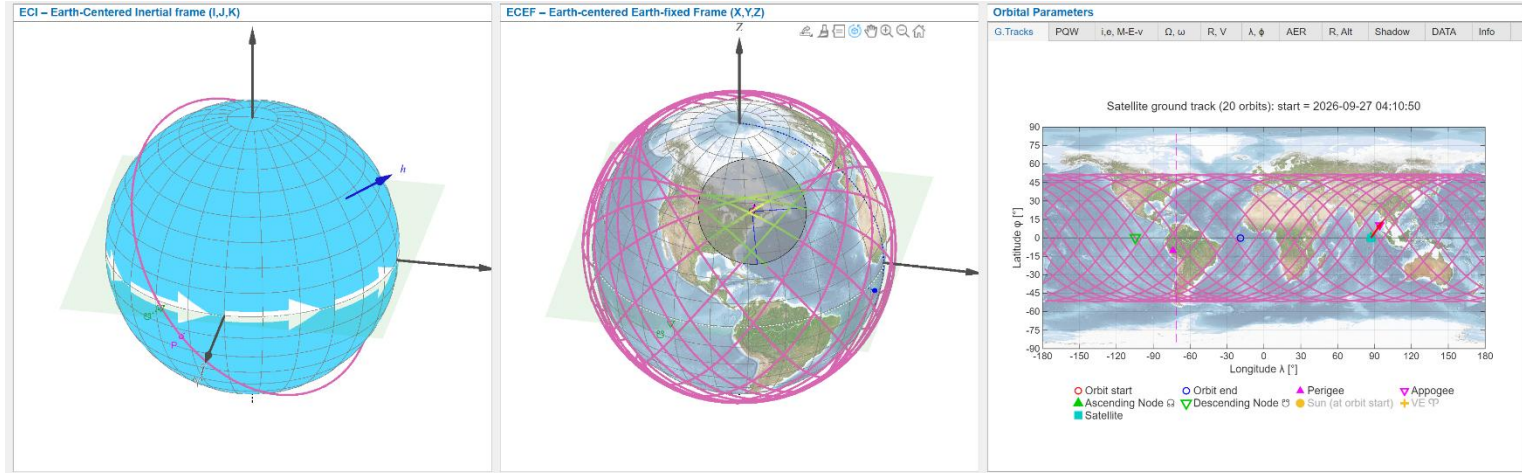
```
0 ISS (ZARYA)
1 25544U 98067A   26270.17419514   .00009528   00000-0   18291-3 0   9997
2 25544   51.6315 155.3455 0007168 193.0560 167.0244 15.48664528587561
```

- You can translate in Python the Extract3LE.m or get a TLE reader from internet

Propagate orbit with 2BK & 2BN methods



- 1a. Write Python code to compute orbit in ECI and ECEF over a number of revolutions, starting from epoch
→ Use approximation $ECEF = Rot_Z(GMST) \times ECI$ transformation
- 1b. Draw orbit in 3D and Ground Track in 2D (equirectangular projection)



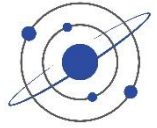
→ Use example code provided in `SOL_3D_Geometry.py` as a starting example

Analyze orbital parameters



2. Draw the variation of anomalies over time
 3. Draw the state vectors over the first orbit
 - Validate the orbital speed with the theoretical value
 4. Draw the satellite altitude above ground
 - Explain behavior (explore by changing orbit eccentricity value)
 5. Draw the variation of latitude λ and longitude φ over time
 - Subsatellite path plotted in the Ground Track figure in equirectangular coordinates
 6. Draw the satellite elevation & range for the observer in Québec (at least 20 orbits) to find maximum overpass time
 - Explain the oscillatory behavior of the curve
 - What is the maximum theoretical elevation possible for this observer latitude and orbit characteristics?
 - What if the observer was located at Iqaluit?
- For all generated figures, give your observations and comments on the physics of the phenomena
- Feel free to use Orbits101 for comparison/validation to your calculations; note that some numerical discrepancies can be observed due to the way ECEF positions are derived from ECI taking into account precession & nutation

Draw SEZ plane with the satellite at maximum elevation



- 7a. Draw in 3D the SEZ plane geometry
- 7b. Draw the vector to the satellite and the subsatellite point
- 7c. Compute the elevation angle with the cross product

